classmate

Date
Page

Activity - 5

Given input {4371, 1323, 6173, 4199, 4344, 9677, 1989} and a hash function $h(x) = x \pmod{10}$, Show the resulting

a) ~~Open~~ seperate chaining hash table

b) open addressing hash table using linear probing

c) Open addressing hash table using quadratic probing

d) Open addressing hash table with record hash function $h_2(x) = 7 - (x \bmod 7)$

Ans Given input {4371, 1323, 6173, 4179, 4344, 9679, 1989}

Hash function : $h(x) = x \bmod 10$

Hash Table size = 10, so indices are 0 to 9

a) Open addressing

key	x mod 10
4371 % 10	1
1323 % 10	3
6173 % 10	3
4199 % 10	9
4344 % 10	4
9679 % 10	9
1989 % 10	9

index

0

1 → 4371

2

3 → 6173 → 1323

4 → 6173

5

6

7

8

9 → 4199 → 9679 → 1989

b) linear hashing

$h(x) = h(\text{key} + i)$

$h(4371) = 4371 \% 10 = 1$

$h(1323) = 1323 \% 10 = 3$

$h(6173) = 6173 \% 10 = 3$

$h(6173, i) = (6173 + 6) = 6173 \% 10 = 3$ p = 1 collision

$(6173+1) = 6174 \% 10 = 4$ p = 0

$h(4199) = 4199 \% 10 = 9$

$h(4344) = 4344 \% 10 = 4$ collision p = 1

$(4344+1) = 4345 \% 10 = 5$ p = 2

$h(9679) \% 10 = 9$ collision

$h(9679, i) = (9679 + 0) = 9679 \% 10 = 9$ p = 2

$= 1989 \% 10 = 9 \quad p = 1$

0	→ 9679
1	→ 4371
2	→ 1989
3	→ 1323
4	→ 6173
5	→ 4344
6	
7	
8	
9	→ 4199

c) open Addressing - Quadratic probing

formula: $h(x,i) = (h(x) + i^2) \bmod 10$

4371 → 1

1323 → 3

6173 → 3 occupied $\quad i = 1 \rightarrow 3+1 = 4$

4199 → 9

4344 → 4 occupied $\quad i = 1 \rightarrow 4+1 = 5$

9679 → 9 occupied $\quad i = 1 \rightarrow 9+1 = 0$

1989 → 9 occupied $\quad i = 1 \rightarrow 0$ occupied

$i = 2 \rightarrow 9+4 = 3$ occupied

$i = 3 \rightarrow 9+9 = 8$

0	→ 9676
1	→ 4371
2	
3	→ 1323
4	→ 6173
5	→ 4344
6	
7	
8	→ 1989
9	→ 4199

1) Double Hashing

$$h_1(x) = x \bmod 10$$
$$h_2(x) = 7 - (x \bmod 7)$$
$$h(x, i) = (h_1(x) + i \times h_2(x)) \bmod 10$$

Key	h_1	h_2
4371	1	4
1323	3	7
6173	3	1
4199	9	7
4344	4	3
9679	9	2
1989	9	6

4371 → 1
1323 → 3
6173 ⇒ 3 occupied
$= 3 + 1(i) = 4$
4199 → 9
4344 → 4 occupied
→ $4 + 1(3) = 7$
9679 → 9 occupied
→ $9 + 2 = 1$ occupied
→ $9 + 4 = 3$ occupied
→ $9 + 6 = 5$
1989 → 9 occupied
→ 5 occupied
1 occupied
7 occupied
3 occupied
repeats

0	
1	→ 4371
2	
3	→ 1323
4	→ 6173
5	→ 9679
6	
7	→ 4344
8	
9	→ 4199